

# 6B Suboi Zenabu first version.docx

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**Submission date:** 18-May-2025 12:28PM (UTC+0200)

**Submission ID:** 2678712149

**File name:** 6B\_Suboi\_Zenabu\_first\_version.docx (13.48K)

**Word count:** 215

**Character count:** 1396

### **Assignment 6B: Abstract (for Conference)**

#### **Title: Understanding the Key drivers of impact of malaria intervention modeling in Ghana**

**Background:** Malaria remains a major public health concern in Ghana, contributing to 40% of outpatient cases. As malaria transmission is highly heterogeneous across the country, a more targeted response is required. One goal outlined in Ghana's current National Malaria Elimination Strategic Plan (NMESP) is to reduce clinical incidence by 50% by 2028. Understanding the key drivers of intervention impact at the district level is crucial to achieving this goal.

**Methods:** To simulate malaria incidence in Ghana, we parameterized the individual-based model OpenMalaria to reflect district-specific intervention combinations from the NMESP. Using variance-based sensitivity analysis, we systematically evaluated the relative importance of key intervention parameters on incidence reduction from 2023 to 2026.

**Results:** While insecticide-treated bednets (ITNs) were generally the most impactful, the sensitivity analysis revealed substantial heterogeneity across districts. In areas with several interventions co-deployed, ITN coverage was the dominant contributor to variability in impact, followed by coverage of indoor residual spraying (IRS), seasonal malaria chemoprevention (SMC), and malaria vaccine.

**Conclusions:** This work highlights the key drivers of impact of malaria interventions and the need for geographically tailored strategies based on these drivers. Future work will expand the sensitivity analysis to include intervention efficacy and geography-specific parameters.

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